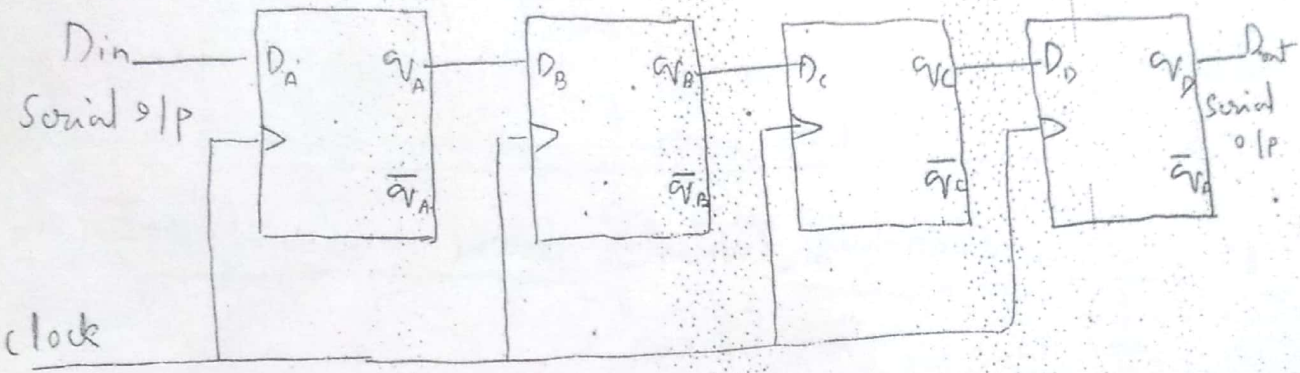
(1) Serial-in-Serial-out (SISO) shift register

The figure below shows the 4-bit SISO shift register using positive edge triggered D flip-flops.



4-bit SISO shift register

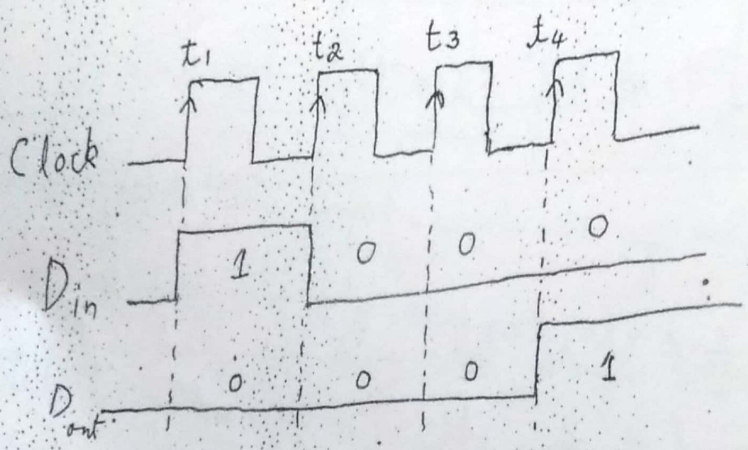
In the case of serial-in-serial-out shift registers input data is fed to the first flip-flop and the o/p is taken from the last flip-flop. The serial input data is available at the output of last flip-flop after the fourth clock pulse. For every clock pulse, the data is shifted to the right from one flip-flop to another. Since the data is shifted in only one direction, they are called as serial-in-serial-out unidirectional shift registers.

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Clock	Serial I/P D_{In}	Serial O/P D_{out}
1	1	0
2	0	0
3	0	0
4	0	1

Function Table of Serial-in-Serial-out

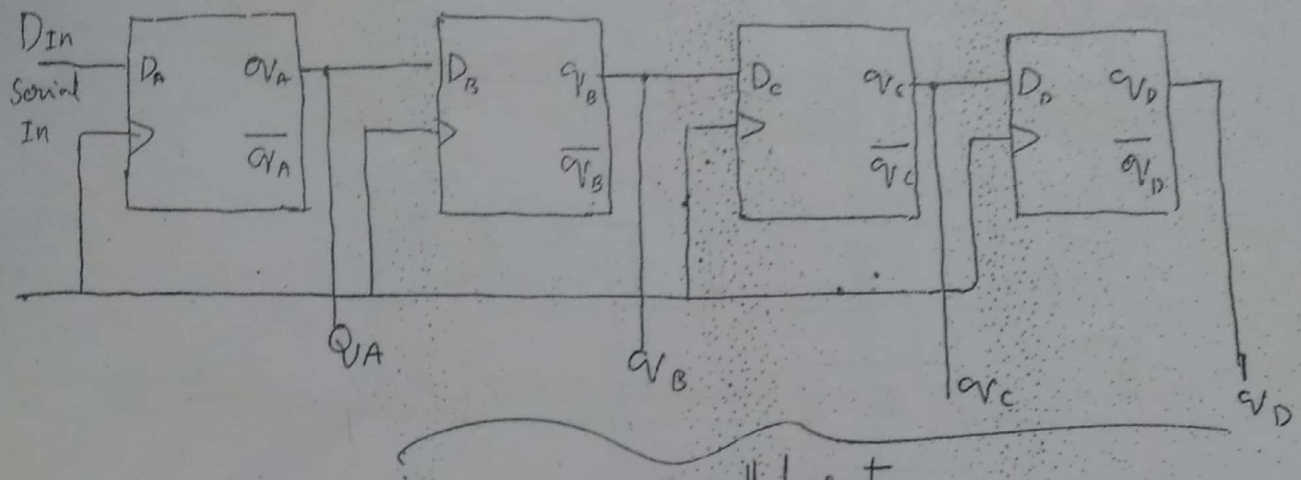


Timing diagram of SISO shift register

Serial-in-parallel-out shift register :- Serial-in-parallel-out using the positive edge triggered D flip-flop is shown below. In case of Serial-in-parallel-out shift register, serial input data is given to the first

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flip-flop and the output is taken simultaneously at all the flip-flop.



4-bit Serial-in-parallel out shift register

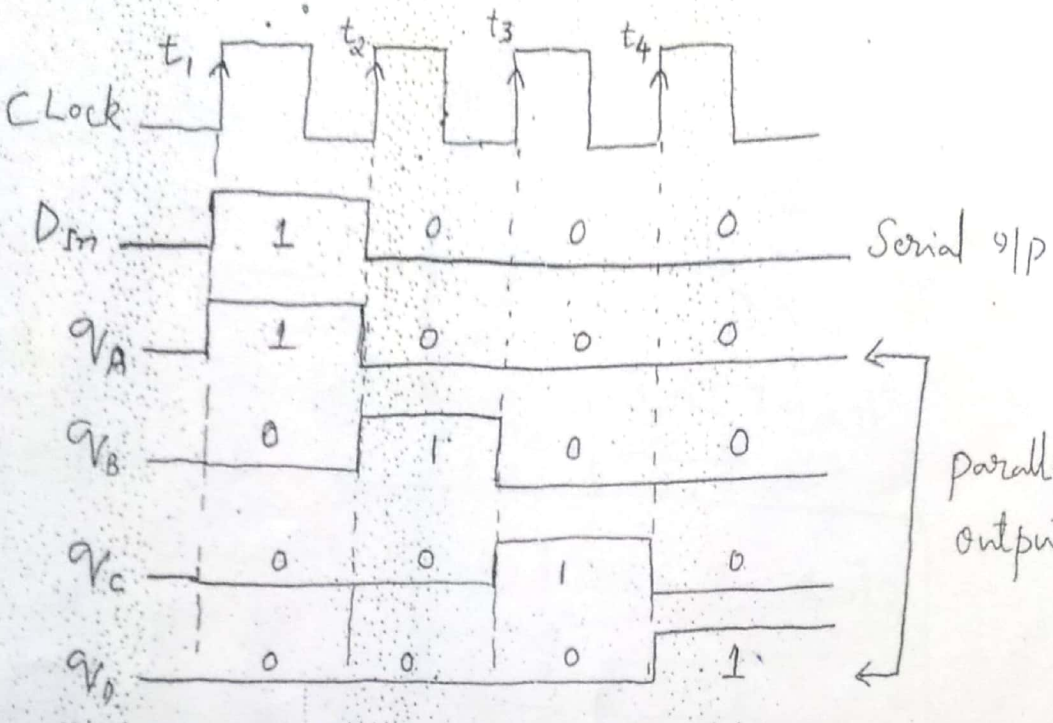
Clock	Serial I/p D_{In}	parallel o/p Q_A	Q_B	Q_C	Q_D
1	1	1	0	0	0
2	0	0	1	0	0
3	0	0	0	1	0
4	0	0	0	0	1

Function Table of Serial-in-parallel-out

Let the serial input data D_{In} = 1000. Upon the

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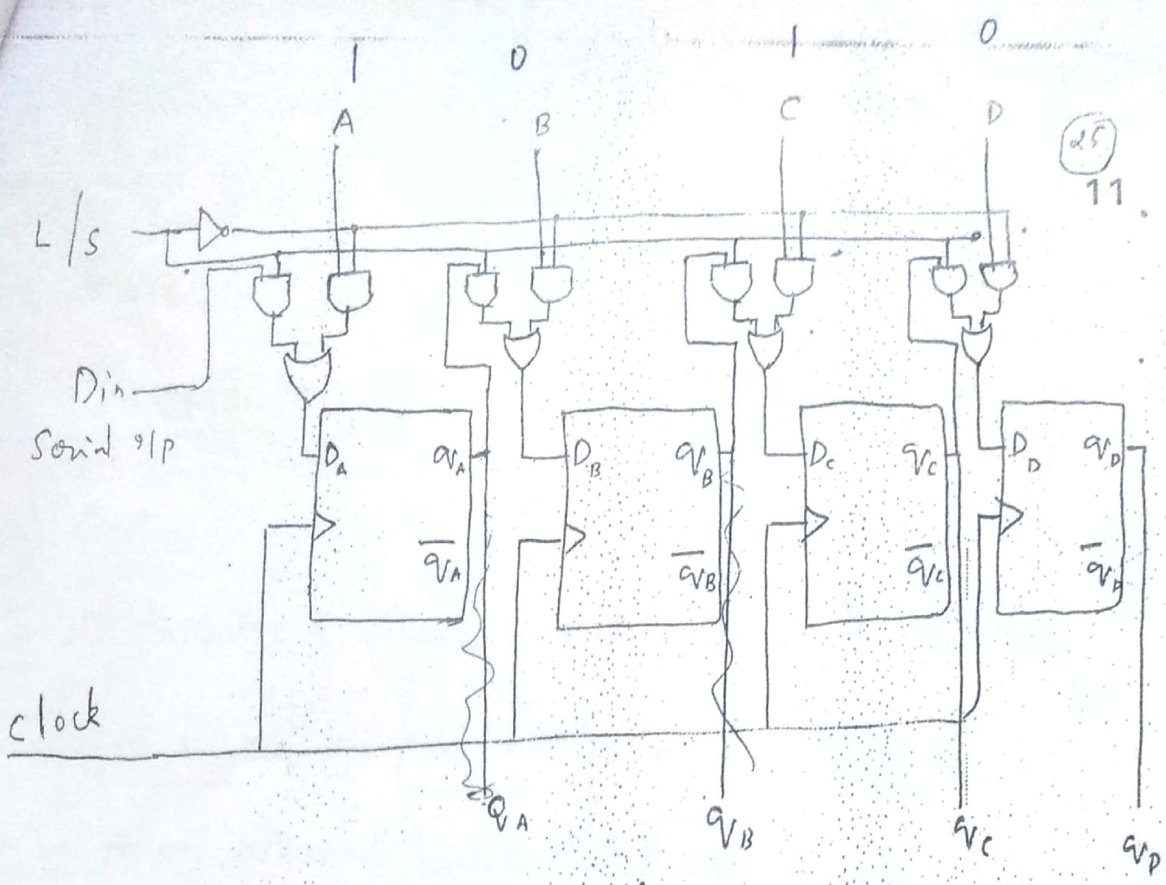
occurrences of the every positive edge of the clock the serial input data shifts from left to right and at the fourth edge of the clock cycle, the data on Q_D becomes 1, i.e., $q_D = 1$.



Timing diagram of 4 bit SIPO shift register

Parallel-in-Serial-out shift register :- In the case of PISO, input data is parallel i.e., input is given to all the flip-flops simultaneously. But the output is taken at the last flip-flop i.e., Serial output.

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Logic diagram of 4 bit Parallel-in-Serial out

The operation of the PISO shift register is controlled by LOAD/SHIFT ($\bar{L}/S$) control line.

When $\bar{L}/S = 0$, the parallel data on the input line ABCD are transferred into the register upon the occurence of the positive edge of clock pulse, because the flip-flop used are the positive edge triggered D flip-flops. When the control line $\bar{L}/S = 1$. it function as a shift register. upon occurence of

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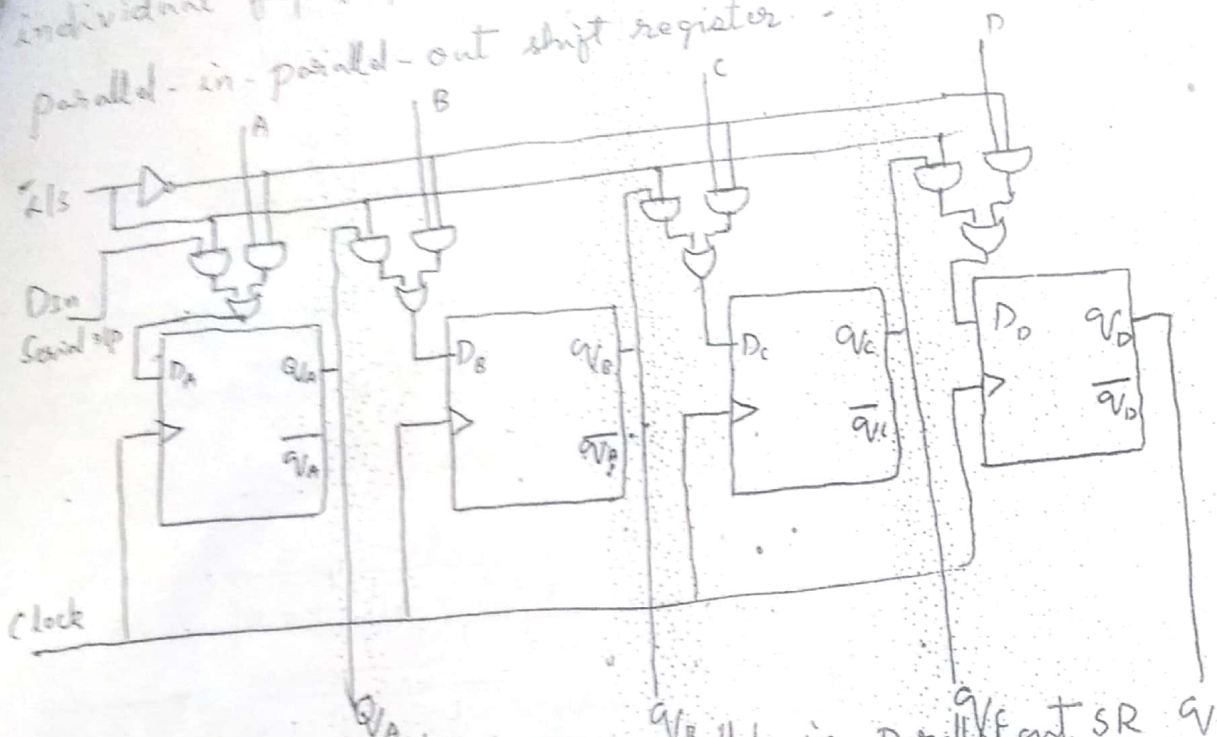
the positive edge of the clock signal data is shifted from one flip-flop to another flip-flop i.e., data is only in the right direction. Hence they are called as in-serial-out unidirectional shift registers.

During the control line $\bar{s}/s=0$, parallel data is loaded into the shift register and during the control line $\bar{s}/s=1$, data is shifted to right and provides serial output. Thus PISO is called parallel to serial Converter. The right shift register is shown above, can be used as a serial-in-serial-out shift register, by applying data at serial input and taking the serial output at Q_D, by keeping the control line $\bar{s}/s=1$.

Parallel-in-parallel-out shift register :- In case of PIPO, the parallel data ABCD is loaded simultaneously upon the occurrence of positive edge clock signal, and at the same time parallel outputs are taken from the

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individual flip-flops simultaneously. Hence they are called parallel-in-parallel-out shift register.

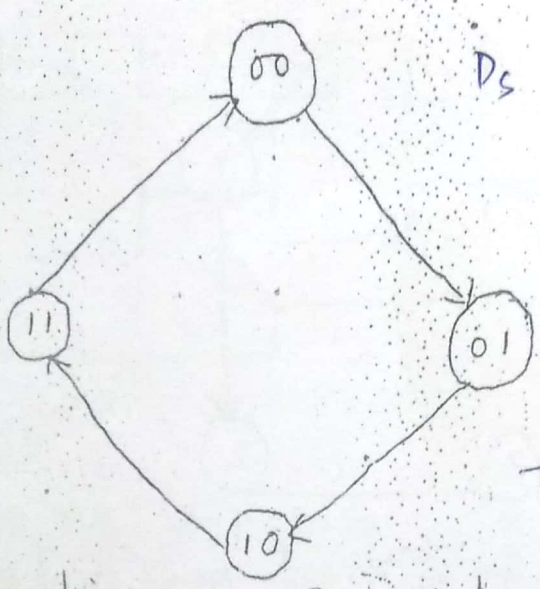


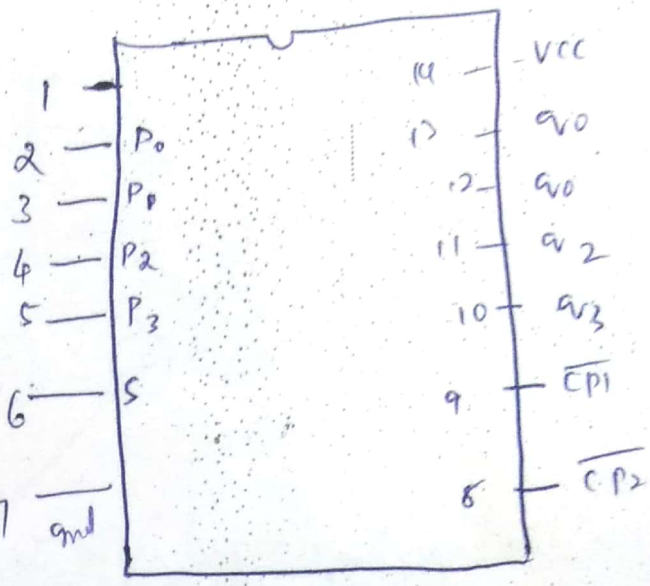
Logic diagram of parallel-in-parallel-out SR

The shift register shown above, can be used as PIPO shift register. The operation of the PIPO shift register is controlled by $\overline{s}/s$, control line. when $\overline{s}/s = 0$, the parallel data on the input line ABCD are transferred into the register upon the occurence of the positive edge of clock pulse, because the flip-flops used are the positive edge triggered D flip-flops. When the control line $\overline{s}/s = 1$, it function as a shift register.

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1) Design a 2 bit. ripple counter using JK flip-flop.



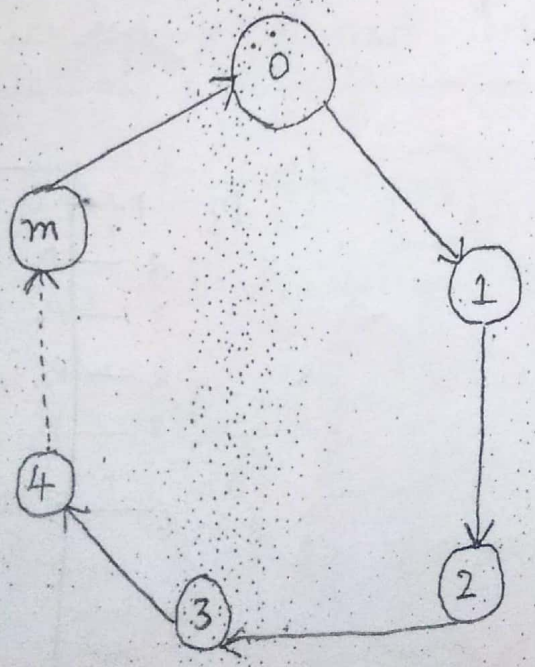


State diagram of mod-4 (2 bit) ripple counter

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Counters :- Counters are the sequential machines, designed to count any required sequence. The basic of a counter is the flip-flop The number of flip-flops required is decided upon the sequence of count. These outputs of the Counters are called as the state of the Counter, and the total number of different states is called its modulus.

The order in which the counting sequence appears is shown by the directed graph is called as the state diagram. The state diagram for mod-m counter is shown below:

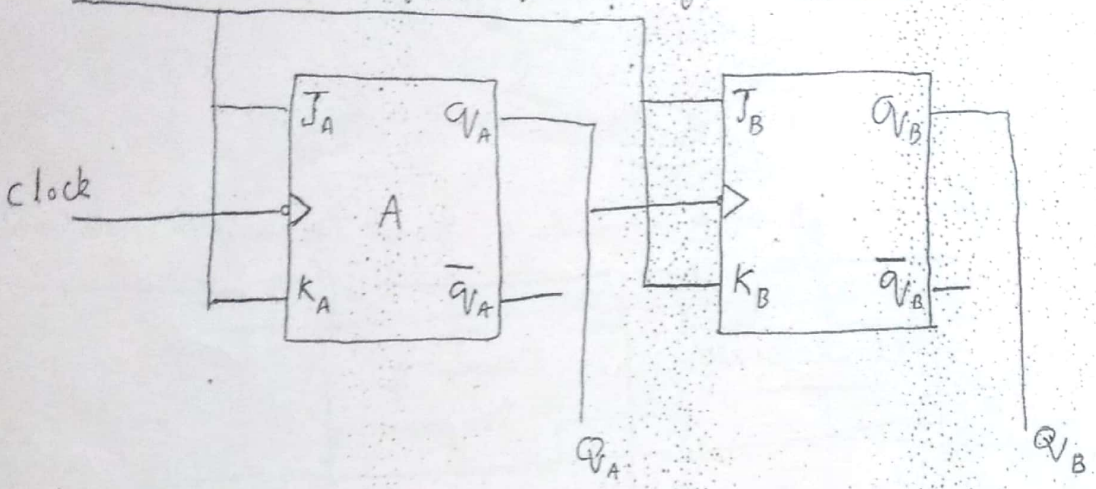


State diagram of a mod-m Counter

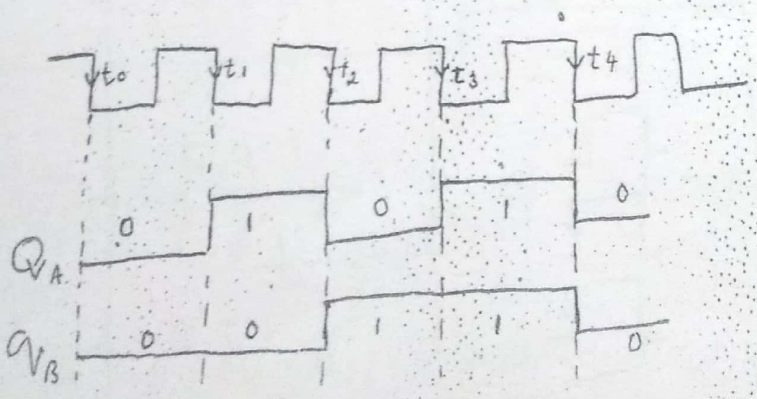
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Clock pulse	Counting Sequence
↓	0 0
↓	0 1
↓	1 0
↓	1 1

Counting sequence of mod-4 Counter



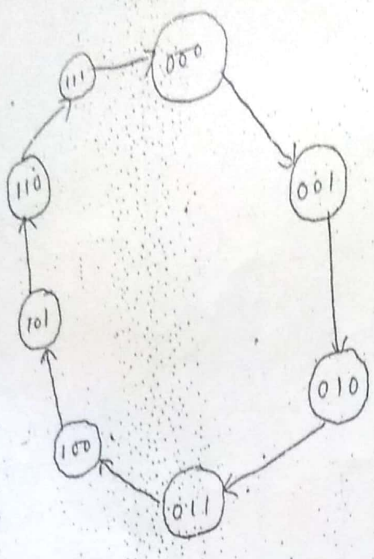
Logic diagram of a 2 bit ripple counter



Timing diagram of a 2 bit ripple counter

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Design a 3 bit Binary Ripple counter using JK ff

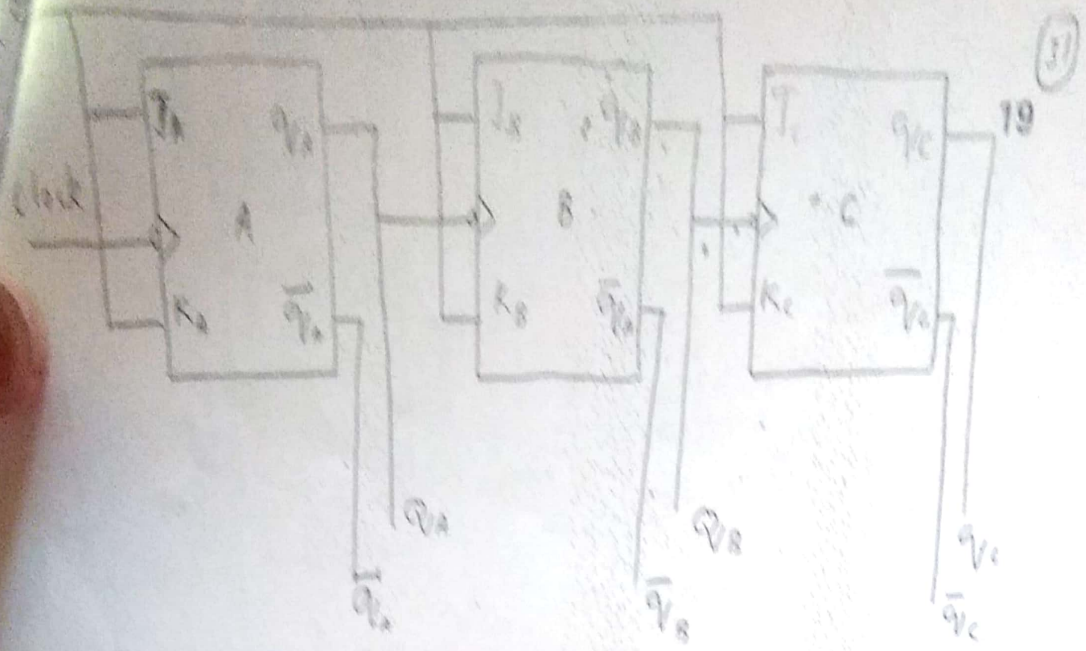


State diagram of mod-8 3-bit Binary Ripple counter

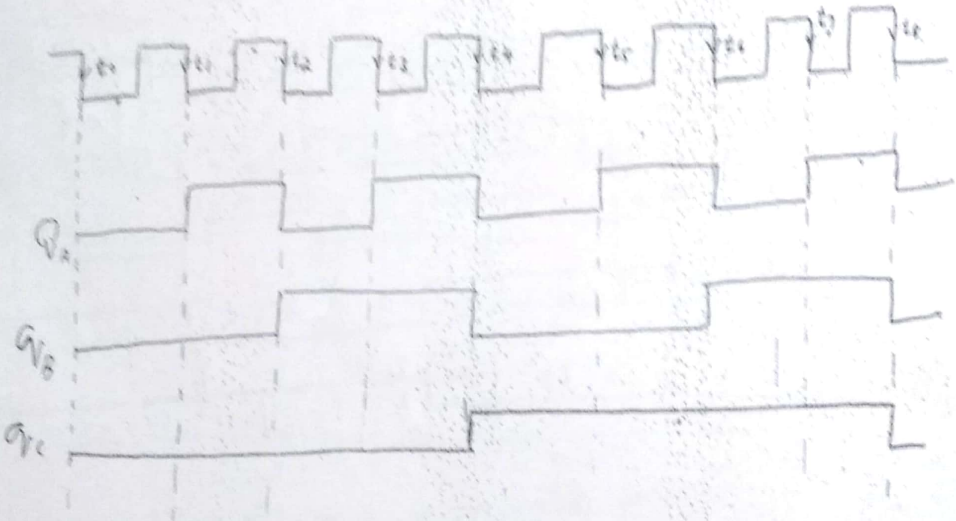
clock pulse	Counting Sequence
	0 0 0
	0 0 1
	0 1 0
	0 1 1
	1 0 0
	1 0 1
	1 1 0
	1 1 1

Counting sequence of mod 8 counter.

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Logic diagram of a 3-bit ripple counter



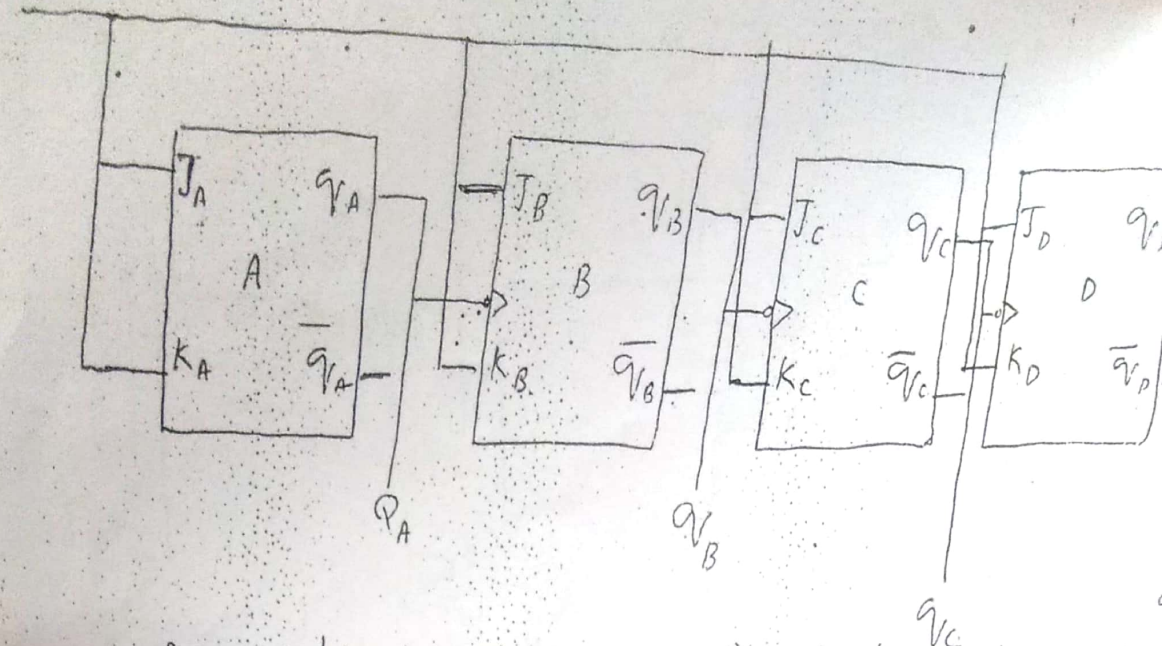
Timing diagram of a 3 bit ripple counter

3) Design a 4 bit Binary Ripple counter using JK flip-flop.

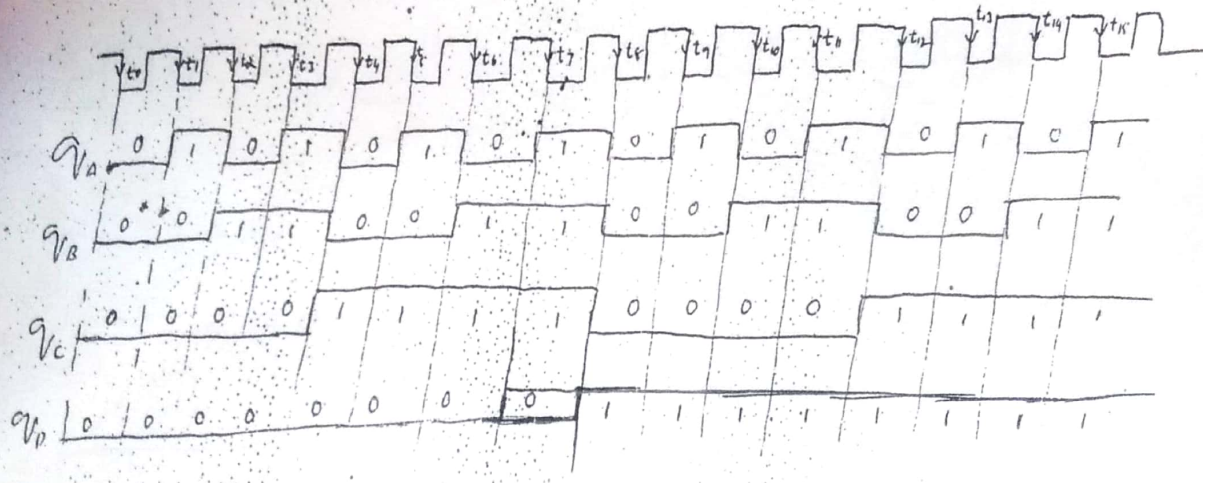
The 4 bit binary Ripple Counter is obtained by adding one

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more flip-flop to the 3-bit binary ripple counter



Logic diagram of a 4-bit ripple counter



Timing diagram of a 4-bit ripple counter

4) Design a Decade Ripple Counter using JK flip-flop.

Decade ripple counter counts from 0000 (0) to 1001 (9), as it is counting from 0 to 9, it is called Decade Counter (BCD counter)

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Design of synchronous Counters :- The design procedure of synchronous Counter involves the following steps

1). The problem is defined by the state diagram or may be the word description or by the Counting sequence.

2) Find the number of flip-flops required and the type of flip-flops

3) Using the excitation table, obtain the transition Table.

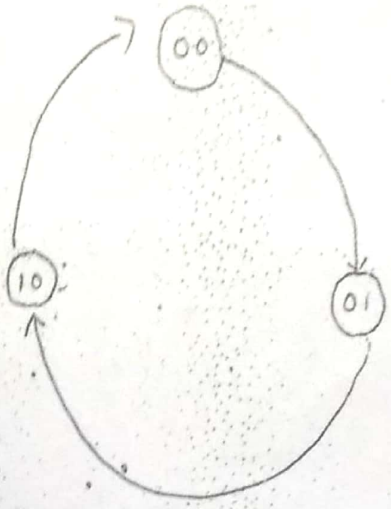
4) Using k-map method or any other simplification method obtain the Boolean Expressions for the flip-flop inputs.

5) Draw the logic diagram.

1) Design Mod-3 synchronous Counter :- The synchronous mod-3 Counter, Counts from 00(0) to 10(2). i.e., 00, 01 and 10, and then repeat the sequence. Since there are three different states, it is called mod-3 Counter. The state

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diagram of mod-3 counter is shown below:



State diagram of mod-3 counter

Steps for the design of mod-3 counter is as follows:

1) problem given is to design mod-3 counter.

2) The number of flip-flops required for mod-3 counter is obtained as follows:

The number of flip-flops = $2^N \geq S$

where $S \longrightarrow$ total number of counting sequence.

$N \rightarrow$ Number of flip-flops required.

For mod-3 counter $S = 3$

$$\therefore 2^N \geq 3$$

with $N = 2$, we have

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$2^2 \geq 3$

$4 \geq 3$

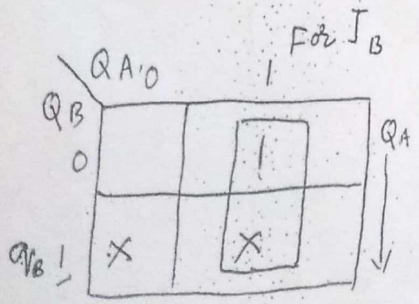
Hence, 4 is greater than 3, the number of flip-flops required are $N = 2$. we use positive edge triggered J-K flip-flop to implement mod-3 counter.

3) Using Excitation table, draw the transition diagram

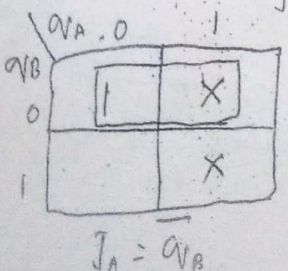
present state		Next state		Flip-flop Inputs			
Q_B	Q_A	Q_{B+1}	Q_{A+1}	J_B	K_B	J_A	K_A
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	0	0	X	1	0	X

state transition table of mod-3 Synchronous counter

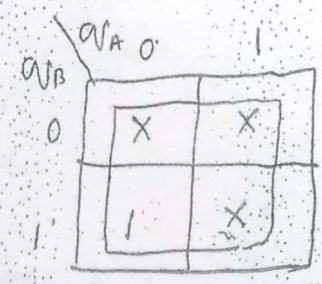
4)



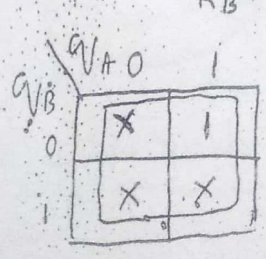
$J_B = Q_A$



$J_A = \overline{Q_B}$

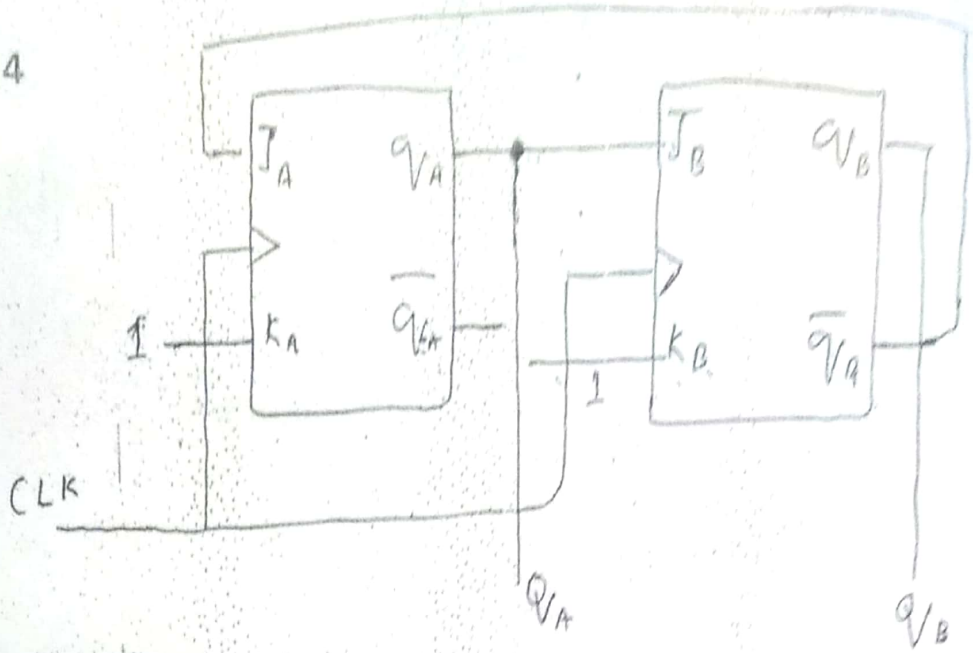


$K_B = 1$ (Vcc)



$K_A = 1$ (Vcc)

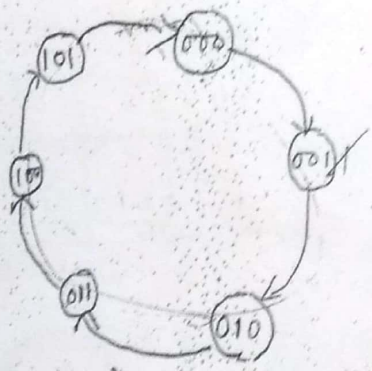
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Logic diagram of mod-3 synchronous counter

2) Design mod-6 synchronous counter using J-K flip-flop.

Step 1 :- Mod-6 counter counts from 000 (0) to 101 (5) and repeats the sequence.



State diagram of mod-6 counter.

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Step 2 :- The number of flip-flop required is

$2^N \geq S$ where $N \rightarrow$ Nos of flip-flop required

$S \rightarrow$ Counting sequence = 6

mod-6

$2^3 \geq 6$

$8 \geq 6$

$N = 3$

$2^m \geq 6$

$2^2 < 6 \quad \therefore = 3$

$8 > 6$

Step 3 :- Using Excitation table, obtain transition state

Present state			Next state			flip-flop input					
						J_C	K_C	J_B	K_B	J_A	K_A
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}						
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1

Step 4 :- using K-map, obtain the boolean Expression for the flip-flop input

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For J_C

Q_C \ $Q_B Q_A$	00	01	11	10
0			1	
1	X	X	X	X

$J_C = Q_B' Q_A$

For K_C

Q_C \ $Q_B Q_A$	00	01	11	10
0	X	X	X	X
1		1	X	X

$K_C = Q_A$

For $J_B = Q_B \bar{Q}_C$

Q_C \ $Q_B Q_A$	00	01	11	10
0		1	X	X
1			X	X

For $K_B = Q_A$

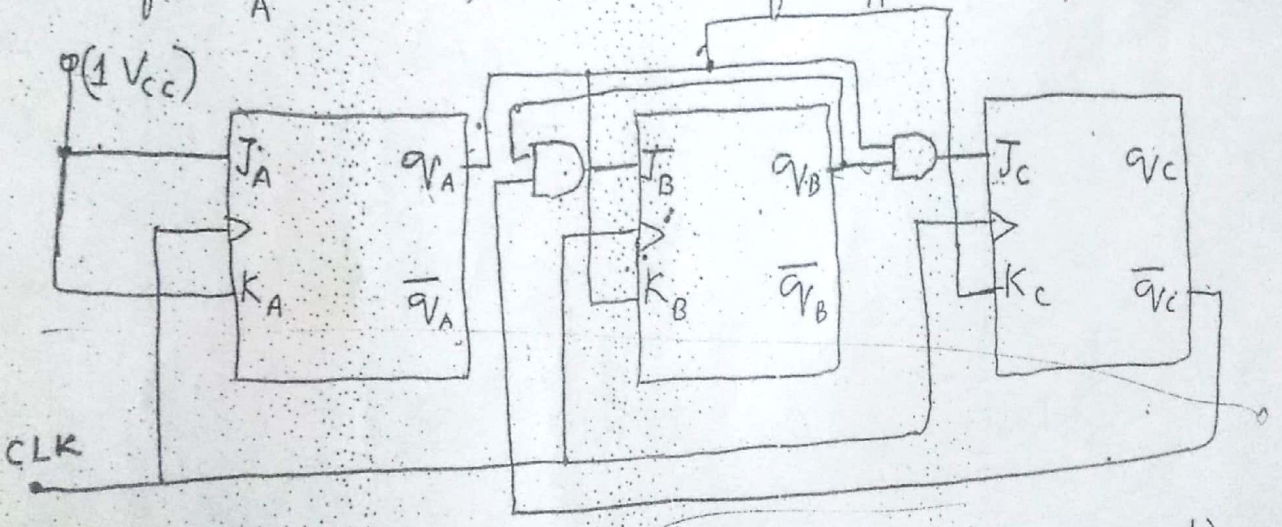
Q_C \ $Q_B Q_A$	00	01	11	10
0	X	X	1	
1	X	X	X	X

For $J_A = 1\ (V_{CC})$

Q_C \ $Q_B Q_A$	00	01	11	10
0	1	X	X	1
1	1	X	X	X

For $K_A = 1\ (V_{CC})$

Q_C \ $Q_B Q_A$	00	01	11	10
0	X	1	1	X
1	X	1	X	X



Mod-6 synchronous Counter using J-K flip-flop

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